values (check that it is irreducible!). The character of this representation is computed similarly to the character of $\mathbb{C}^4$, or from the orthogonality formula.

4.9. Computing tensor product multiplicities using character tables

Character tables allow us to compute the tensor product multiplicities N_{ij}^k using

$$V_i \otimes V_j = \sum N_{ij}^k V_k, \quad N_{ij}^k = (\chi_i \chi_j, \chi_k).$$

Example 4.9.1. The following tables represent computed tensor product multiplicities of irreducible representations of S_3, S_4, and A_5, respectively:

S_3	$\mathbb{C}_+$	$\mathbb{C}_-$	$\mathbb{C}^2$
$\mathbb{C}_+$	$\mathbb{C}_+$	$\mathbb{C}_-$	$\mathbb{C}^2$
$\mathbb{C}_-$		$\mathbb{C}_+$	$\mathbb{C}^2$
$\mathbb{C}^2$			$\mathbb{C}_+ \oplus \mathbb{C}_- \oplus \mathbb{C}^2$

S_4	$\mathbb{C}_+$	$\mathbb{C}_-$	$\mathbb{C}^2$	$\mathbb{C}_+^3$	$\mathbb{C}_-^3$
$\mathbb{C}_+$	$\mathbb{C}_+$	$\mathbb{C}_-$	$\mathbb{C}^2$	$\mathbb{C}_+^3$	$\mathbb{C}_-^3$
$\mathbb{C}_-$		$\mathbb{C}_+$	$\mathbb{C}^2$	$\mathbb{C}_-^3$	$\mathbb{C}_+^3$
$\mathbb{C}^2$			$\mathbb{C}_+ \oplus \mathbb{C}_- \oplus \mathbb{C}^2$	$\mathbb{C}_+^3 \oplus \mathbb{C}_-^3$	$\mathbb{C}_+^3 \oplus \mathbb{C}_-^3$
$\mathbb{C}_+^3$				$\mathbb{C}_+ \oplus \mathbb{C}^2 \oplus \mathbb{C}_+^3 \oplus \mathbb{C}_-^3$	$\mathbb{C}_- \oplus \mathbb{C}^2 \oplus \mathbb{C}_+^3 \oplus \mathbb{C}_-^3$
$\mathbb{C}_-^3$					$\mathbb{C}_+ \oplus \mathbb{C}^2 \oplus \mathbb{C}_+^3 \oplus \mathbb{C}_-^3$

A_5	$\mathbb{C}$	$\mathbb{C}_+^3$	$\mathbb{C}_-^3$	$\mathbb{C}^4$	$\mathbb{C}^5$
$\mathbb{C}$	$\mathbb{C}$	$\mathbb{C}_3^+$	$\mathbb{C}_-^3$	$\mathbb{C}^4$	$\mathbb{C}^5$
$\mathbb{C}_+^3$		$\mathbb{C} \oplus \mathbb{C}^5 \oplus \mathbb{C}_+^3$	$\mathbb{C}^4 \oplus \mathbb{C}^5$	$\mathbb{C}_-^3 \oplus \mathbb{C}^4 \oplus \mathbb{C}^5$	$\mathbb{C}_+^3 \oplus \mathbb{C}_-^3 \oplus \mathbb{C}^4 \oplus \mathbb{C}^5$
$\mathbb{C}_-^3$			$\mathbb{C} \oplus \mathbb{C}^5 \oplus \mathbb{C}_+^3$	$\mathbb{C}_+^3 \oplus \mathbb{C}^4 \oplus \mathbb{C}^5$	$\mathbb{C}_+^3 \oplus \mathbb{C}_-^3 \oplus \mathbb{C}^4 \oplus \mathbb{C}^5$
$\mathbb{C}^4$				$\mathbb{C}_+^3 \oplus \mathbb{C}_-^3 \oplus \mathbb{C}$ $\oplus \mathbb{C}^4 \oplus \mathbb{C}^5$	$\mathbb{C}_+^3 \oplus \mathbb{C}_-^3 \oplus 2\mathbb{C}^5 \oplus \mathbb{C}^4$
$\mathbb{C}^5$					$\mathbb{C} \oplus \mathbb{C}_+^3 \oplus \mathbb{C}_-^3$ $\oplus 2\mathbb{C}^4 \oplus 2\mathbb{C}^5$